

RobbieIN THE UNITED STATES PATENT AND TRADEMARK OFFICE

In re Application of:
Robert Ernest Troxler

Confirmation No.: 3345

Application No.: 17/167,404

Art Unit: 3992

Filing Date: February 4, 2024

Examiner: MENEFEE, JAMES A

**For: SYSTEMS AND METHODS FOR HIGH VOLTAGE CONVERSION AND
MULTIPLICATION FOR IONIZING RADIATION DETECTION**

Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

Examiner:

RESPONSE TO OFFICE ACTION

In reply to the Office Action dated May 8, 2025, please consider the following:

Amendments to the Claims as reflected in the listing of the claims; and

Remarks.

This listing of claims will replace all prior versions, and listings, of claims in the application.

Listing of Claims:

1. (Currently Amended) A neutron detector comprising:
at least one detector configured for detecting ionizing radiation; wherein the at least one detector is [a neutron absorbing] coated in an active region interior of a neutron absorbing gas tube; and
the interior of the at least one gas tube is coated with a predetermined mass thickness of [optimally of about 1] 0.75-1.25 mg/cm² in Boron 10; wherein the detector gas contains a predetermined mixture of Neon, Argon, a [substantially] reduced amount of a neutron absorbing gas component including partial pressures of at least one of ³He and [BF³] BF₃ and a quench component;
whereas the argon, neon and the quench component [components] increase the total pressure and [improve] adjust the waveform of the detector output to form for measurement.
2. (Original) The neutron detector of claim 1, wherein the interior of the at least one gas tube is coated in a substantially pure Boron-10 and also comprises a neutron absorbing gas and a Penning gas for adjusting the pressure and electronic waveform.
3. (Currently Amended) The neutron detector of claim 1, wherein the interior of the at least one gas tube is coated in a substantially Boron-10 and also comprises a mix of a Penning gas and a BF₃ [BF³] gas whereas the BF₃ [BF³] gas is less than 700 Torr in pressure for absorbing thermal neutrons and the penning gas adjusts the waveform signal and gain.
4. (Currently Amended) The neutron detector of claim 1, wherein the interior of the at least one gas tube is coated with Boron-10 nanoparticles and also comprises a mix of a Penning gas and a [BF³] BF₃ gas less than 700 Torr for adjusting the waveform signal and efficiency.
5. (Currently Amended) The neutron detector of claim 1, wherein the interior of the at least one gas tube is coated with Boron-10 nanoparticles and also comprises a mix of a

Penning gas and at least one of a reduced amount of ^3He and a $[\text{BF}_3]$ BF_3 gas for adjusting the waveform signal and efficiency; whereas the pressure is reduced below 700 Torr.

6. (Currently Amended) The neutron detector of claim 1, wherein [the] a cathode of the at least one gas tube contains an alloy of Boron with a non-corrosive metal and further comprises a mix of a Penning gas and a $[\text{BF}_3]$ BF_3 gas of less than 700 Torr for adjusting the waveform signal and efficiency.

7. (Currently Amended) The neutron detector of claim 1, wherein the at least one gas tube is a straw detector of Boron and further comprises a mix of a Penning gas and at least one of a ^3He and $[\text{BF}_3]$ BF_3 gas of less than 700T for adjusting the waveform signal and efficiency.

8. (New) A hybrid neutron detector comprising: at least one detector configured for detecting ionizing radiation; wherein the at least one detector comprises a neutron absorbing gas tube; and the active interior of the at least one gas tube coated optimally with about $1\text{mg}/\text{cm}^2$ in Boron 10 or a neutron interacting component material; wherein the detector further contains a mixture of at least one of a non-neutron interacting fill gas, gas, and/or a quench component, and a reduced amount of a neutron interacting gas component including partial pressures of at least one of ^3He and BF_3 ;

whereas the fill, quench and non-neutron interacting gas components increase the total pressure and adjust the waveform of the detector output.

9. (New) The neutron detector of claim 8, wherein the interior of the at least one gas tube is coated in an enriched Boron-10 and also comprises a neutron interacting gas and at least one of Ar, Ne, ^4He , Br, CO_2 or a Penning gas for adjusting the pressure and electronic waveform.

10. (New) The neutron detector of claim 8, wherein the interior of the at least one gas tube is coated in an enriched Boron-10 and also comprises a fill mix of at least one of Ar, Ne,

CO₂, ⁴He, Br or Penning gas and a BF₃ gas whereas the BF₃ gas is less than 700 Torr in partial pressure for interacting with thermal neutrons and the fill gas adjusts the waveform signal and gain.

11. (New) The neutron detector of claim 8, wherein the interior of the at least one gas tube is coated with at least one of a pure neutron interacting material including Boron-10 nanoparticles, nano powders, spherical nanoparticles, nanoparticles, Boron nitride, Crystalline 10Boron, 10B Enriched Boron carbide, 10B enriched metal, Boride ceramics/Metal Borides and 10B enriched Boron Trifluoride and conformal coatings, and further comprises a mix of a non-interacting fill and quench gas and a neutron interaction gas for adjusting the waveform signal and efficiency.

12. (New) The neutron detector of claim 8, wherein the interior of the at least one gas tube is coated with Boron-10 nanoparticles and also comprises a mix of at least one of Ar, Ne, a Penning gas and at least one of a reduced amount of ³He and a BF₃ gas for adjusting the waveform signal and efficiency; whereas the BF₃ partial pressure is reduced below one atmosphere.

13. (New) The neutron detector of claim 8, wherein the cathode of the at least one gas tube is one of an alloy or composite of Boron with a non-corrosive metal and further comprises a mix of at least one of Ar, Ne, and Penning gas and a neutron interacting gas for adjusting the waveform signal and efficiency.

14. (New) The neutron detector of claim 8, wherein the at least one gas tube interior is at least one of a straw detector, lined tubes, or a cathode coating of Boron and further comprises a mix of at least one of Ar, Ne, and a Penning gas with at least one of a ³He and BF₃ gas for adjusting the waveform signal and efficiency.

15. (New) A neutron detector comprising:
at least one detector configured for detecting ionizing radiation; wherein the at least

one detector includes a neutron interacting gas tube; and

the interior of the at least one gas detector is further coated in a neutron interacting material up to 1 mg/cm² of a Boron 10 or neutron capturing component material; wherein the detector gas contains partial pressures of an ionizing gas and 3% to 93% amount of a neutron interacting gas component including partial pressures of at least one of ³He and BF₃ and a quench component; whereas the ionization and quench components increase the total pressure and adjust the waveform of the detector output.

16. (New) The neutron detector of claim 15, wherein the ionizing gas comprises at least one of Ar, Ne, Br, Halogens, CO₂, Br, and a Penning gas.

17. (New) The neutron detector of claim 15, wherein the quench component comprises at least one of a Halogen gas, CO₂ gas, active and a passive RC or external circuit.

18. (New) The neutron detector of claim 15, wherein the at least one gas tube is coated with about 1mg/cm³ or lined in B10 or made up of neutron interacting straws.

19. (New) The neutron detector of claim 15, wherein the at least one gas tube is coated with at least one of Boron-10, Crystalline Boron 10, Boron-10 Enriched Boron carbide, Boron 10 enriched metal, Boride ceramics/Metal Borides and Li-6.

20. (New) The neutron detector of claim 15, wherein the Boron-10 is enriched Boron Trifluoride, Crystalline ¹⁰Boron, ¹⁰B Enriched Boron carbide, ¹⁰B enriched metal, Boride ceramics/Metal Borides.

21. (New) The neutron detector of claim 15, wherein the final partial pressures sum to less than about 70 psi of ³H₃ or BF₃.

22. (New) The neutron detector of claim 15, wherein the partial pressures sum to 70 pounds per square inch (PSI) or from less than 1 atmosphere to about 5 atmospheres of ³H₃ or

less than 1 atmosphere of BF_3 .

23. (New) The neutron detector of claim 15, wherein the cathode is boron carbide/aluminum matrix.

24. (New) A neutron detector comprising:
at least one detector configured for detecting ionizing radiation; wherein the at least one detector is coated in an active region interior of a neutron interacting gas tube; and
the interior of the at least one gas tube contains at least one of neutron capturing coated straws, coated cathodes, nanoparticles, nanopowders, and Boron Nitride, increasing the active surface area of a neutron interacting material; wherein the detector gas contains a mixture of an ionizing gas, and a substantially reduced amount of a neutron interacting gas components including partial pressures of at least one of ^3He and BF_3 whereas the ionization gas components and pressures include at least one of Argon, Neon, Penning, CO_2 , Br, ^4He for a total pressure from less than 1 atmosphere up to the total pressure of 70 Psi.

25. (New) The neutron detector of claim 24, wherein the mixture is tuned for parameters of count rate, starting voltage, breakdown, output pulse voltage, and plateau shape.

26. (New) The neutron detector of claim 24, wherein the at least one detector is a neutron interacting Boron based gas tube.

27. (New) The neutron detector of claim 24, wherein the interior of the at least one Boron gas tube is coated in a substantially pure Boron-10 coating Crystalline ^{10}B , ^{10}B Enriched Boron carbide, ^{10}B enriched metal, Boride ceramics/Metal Borides.

28. (New) The neutron detector of claim 24, wherein the interior of the at least one Boron gas tube is coated in a substantially pure Boron-10 coating and also comprises a Penning gas for adjusting output parameters.

29. (New) The neutron detector of claim 24, wherein the interior of the at least one Boron gas tube is coated with a range of 0.75mg/cm² to 1.25mg/cm² of substantially pure Boron-10, a range of at least one of about 5cmHg-70cmHg partial pressure of BF₃ and 70 psi ³He of an interacting gas, and less than 1 atmosphere of a non-neutron interacting fill gas and quench gas.

30. (New) The neutron detector of claim 24, wherein the coating is applied through a thick film processes including painting, or more controlled and cleaner processes, as used in thin film technology such as sputtering, CVD, Plasma enhanced CVD, evaporation, magnetron sputtering, glow discharge processes, and conformal coatings.

31. (New) The neutron detector of claim 24, wherein the at least one Boron gas tube comprises a mix of at least one of helium, boron trifluoride, argon, neon, carbon dioxide, Penning gas, and a halogen gas.

32. (New) The neutron detector of claim 31, wherein the detector contains a neutron interacting material coating, a gas mix, and a mix which has no specific interaction with a neutron, whereas the interacting material contains pure Boron 10 and the interacting gas mix contains at least one of a BF₃ and He₃; furthermore the non-interacting gas mix contains at least one of CO₂, Argon, Neon, Br, Penning, an ionizing gas and a quench mechanism.

33. (New) A neutron detector comprising:
at least one detector configured for detecting ionizing radiation; wherein the at least one detector is a neutron absorbing gas tube; and
the interior of the at least one gas tube is coated optimally of about 1 mg/cm² in Boron 10; wherein the detector gas contains a mixture of Neon, Argon, a reduced amount of a neutron absorbing gas component including partial pressures of at least ³He and BF₃ and a quench component;
whereas the argon, neon and quench components increase the total pressure and adjust the waveform of the detector output.

34. (New) The neutron detector of claim 33, wherein the interior of the at least one gas tube is coated in a substantially pure Boron 10 and also comprises a neutron absorbing gas and a Penning gas for adjusting the pressure and electronic waveform.
35. (New) The neutron detector of claim 33, wherein the interior of the at least one gas tube is coated in a substantially Boron-10 and also comprises a mix of a Penning gas and a BF₃ gas whereas the BF₃ gas is less than 700 Torr in pressure for absorbing thermal neutrons and the penning gas adjusts the waveform signal and gain.
36. (New) The neutron detector of claim 33, wherein the interior of the at least one gas tube is coated with Boron-10 nanoparticles and also comprises a mix of a Penning gas and a BF₃ gas less than 700 Torr for adjusting the waveform signal and efficiency.
37. (New) The neutron detector of claim 33, wherein the interior of the at least one gas tube is coated with Boron-10 nanoparticles and also comprises a mix of a Penning gas and at least one of a reduced amount of ³He and a BF₃ gas for adjusting the waveform signal and efficiency; whereas the pressure is reduced below 700 Torr.
38. (New) The neutron detector of claim 33, wherein the cathode of the at least one gas tube contains an alloy of Boron with a non-corrosive metal and further comprises a mix of a Penning gas and a BF₃ gas of less than 700 Torr for adjusting the waveform signal and efficiency.
39. (New) The neutron detector of claim 33, wherein the at least one gas tube is a straw detector of Boron and further comprises a mix of a Penning gas and at least one of a ³He and BF₃ gas of less than 700T for adjusting the waveform signal and efficiency.
40. (New) The neutron detector of claim 1, wherein the at least one detector is coated of about 1mg/cm² of the neutron absorbing gas tube.

41. (New) The neutron detector of claim 1, further comprising one or more temperature sensors configured to acquire temperature data for correction processes.

REMARKS

Claims 1-32 are pending in this application. By this response, various claims are amended, and new claims are added. No new matter has been added.

Favorable consideration and withdrawal of the rejection in light of the amendments and remarks are respectfully requested.

I. Claim Objections

Claims 1, 6, and 8-32 stand objected to as not being properly marked. It is respectfully submitted that the presented submitted claim set is correctly marked, and that the objection should be withdrawn. This objection is respectfully traversed.

Regarding Claim 1, the issue with “3HE” and “BF3” set forth on Page 3 is believed to be fully addressed by Claim 1 presented herein.

Regarding Claim 6, the issue with “BF3” set forth on Page 3 is believed to be fully addressed by Claim 6 presented herein.

Regarding Claims 8-41, the new claims set forth herein are completed underlined. Therefore, it is believed that the requirements are met by this submission.

Further, the Office Action references Claim 33 and the use of superscripts when subscripts should be use for “BF3”. The claims are presented and amended in the claims submitted herewith to use subscript “3” for “BF₃”.

Regarding Claims 8-32, these new claims address the aforementioned issued.

In view of the foregoing, it is respectfully submitted that the objection to the claims should be withdrawn.

II. Claim Rejections Under 35 USC Section 251

Claims 15-28 and 30 stand rejected under 35 U.S.C. § 251 as presenting new matter. Particularly, the Office Action contends that Claims 15-28 and 30 are improper as recapturing broadened claim subject matter surrendered in the application for the patent upon which the present reissue is based. It is believed that the independent claims among Claims 15-28 and 30 are amended such that they do not recapture broadened claim subject matter. Accordingly, it is respectfully submitted that the rejection should be withdrawn.

III. **Claim Rejections Under 35 U.S.C. § 112**

35 U.S.C. § 112(a) or 35 U.S.C. § 112, First paragraph

Claims 15-30, 33-39, and 41 stand rejected under 35 U.S.C. § 112(a) or 35 U.S.C. § 112, first paragraph, under the contention that the claims do not comply with the written description requirement. It is believed that Claims 15-30, 33-39, and 41 are amended such that they comply with the written description requirement. Accordingly, it is respectfully submitted that the rejection should be withdrawn.

Further, example support the new claims and claim amendments are set forth in the following table:

Claim	Example Support in U.S. Patent No. 10,197,686
1	Column 4, Lines 25-40
2	Same
3	Same
4	Same
5	Same
6	Same
7	Same
8	Column 4, Lines 25-40; Column 8, Lines 48-51
9	Column 9, Lines 35-38; Column 8, Lines 55-67
10	Same
11	Column 9, Lines 35-38
12	Column 5, Lines 53 and 54
13	Same
14	Column 4, Lines 1-5; Column 5, Line 54
15	Column 4, Lines 25-40
16	Column 4, Lines 15-20
17	Column 4, Lines 15-20, and 25-40
18	Column 9, Lines 2 and 27-37
19	Column 9, Lines 27-37
20	Column 5, Line 56; Column 9, Lines 17 and 36-38
21	Same
22	Column 9, Lines 27-38
23	Column 5, Line 56
24	Column 4, Lines 25-40
25	Column 4, Lines 58-65
26	Same

27	Column 5, Line 56; Column 9, Lines 36-38; Column 9, Line 17
28	Same
29	Column 4, Lines 25-40
30	Column 9, Lines 1-16
31	Same
32	Same
33	Column 4, Lines 25-40
34	Same
35	Same
36	Column 9, Lines 8-16
37	Same
38	Same
39	Same
40	Same
41	Column 6, Lines 59-61

35 U.S.C. § 112(b) or 35 U.S.C. § 112, Second Paragraph

Claims 1-41 stand rejected under 35 U.S.C. § 112(b) or 35 U.S.C. § 112, second paragraph, under the contention that the claims are indefinite for not particularly pointing out and distinctly claiming the subject matter regarded as the invention. This rejection is respectfully traversed.

Regarding Claim 1, the Office Action sets forth issues with “substantially reduced amount of neutron absorbing gas component”. Without conceding the merits of the contentions and merely to advance prosecution, the term “substantially” is removed from the phrase to render to rejection of Claim 1 moot.

Further regarding Claim 1, the claim recites “quench components” in the second to last line, whereas “a quench component” is referenced earlier in the claim. In the amendment to Claim 1 herein, “quench components” is replaced with “the quench component”. Therefore, it is believed that the antecedent basis issue is fully addressed.

Regarding Claim 6, the Office Actions states that the feature “the cathode” lacks antecedent basis. The claim is amendment to remove “the”. Therefore, it is believed that the antecedent basis issue is fully addressed.

It is believed that new Claims 8-41 particularly point out and distinctly claim the subject matter regarded as the invention.

35 U.S.C. § 112(d) or 35 U.S.C. § 112, Fourth Paragraph

Claims 31, 37, 39, and 40 stand rejected under 35 U.S.C. § 112(d) or 35 U.S.C. § 112, fourth paragraph, under the contention that the claims are indefinite for not particularly pointing out and distinctly claiming the subject matter regarded as the invention. It is believed that new Claims 8-41 particularly point out and distinctly claim the subject matter regarded as the invention. Accordingly, it is respectfully submitted that the rejection should be withdrawn.

IV. Claim Rejections Under 35 U.S.C. § 103

Claims 1-3, 6-10, 13-19, 21-32, and 38-40 stand rejected under 35 U.S.C. § 103 under the contention that the claims are unpatentable over U.S. Patent Application Publication No. 2011/0114848 to Frank

et al. (hereinafter, “Frank”) in view of U.S. Patent Application Publication No. 2014/0117246 to Zhou et al. (hereinafter, “Zhou”) and further in view of U.S. Patent Application Publication No. 2005/0258373 to Lacy (hereinafter, “Lacy”). Further, Claims 4, 5, 11, 12, 36, and 37 stand rejected as being unpatentable over Frank, Zhou, and Lacy and further in view of U.S. Patent Application Publication No. 2011/0284755 to Stradins et al. (hereinafter, “Stradins”). Claim 20 stand rejected under 35 U.S.C. § 103 under the contention that the claim is unpatentable over Frank, Zhou, and Lacy and further in view of U.S. Patent No. 8,384,004 to Rowland et al. (hereinafter, “Rowland”). These rejections are respectfully traversed.

Frank does not teach or fairly suggest the features of amended Claim 1, Claim 8, Claim 15, Claim 24, or Claim 33. Particularly, for example, Frank does not teach or suggest the amended Claim 1 feature of the interior of the at least one gas tube is coated with a predetermined mass thickness of 0.75-1.25 mg/cm² in Boron 10; wherein the detector gas contains a predetermined mixture of Neon, Argon, a reduced amount of a neutron absorbing gas component including partial pressures of at least one of ³He and BF₃ and a quench component; and whereas the argon, neon and the quench component increase the total pressure and adjust the waveform of the detector output to form for measurement. In the Abstract, it is

described that GM tubes detecting gamma rays with reactive gases. Extending the life of a detector including an anode resistant to erosion caused by the internal detector gases.

Paragraph [0027] of Frank discusses that for a neutron tube with He3 or BF3. The problem here is that He3 tubes use inert gases, and these do not affect the anode over life. In this paragraph, the manner of instantly mentioning “neutron tubes” Frank includes in this sentence conversion coatings too. Nothing is said about solid coatings that absorb neutrons and emit charged particles, and nothing is said about increasing the efficiency of neutron tubes by a hybrid approach of neutron absorbing coatings with a reduced amount of neutron absorbing gas. Frank is simply listing different types of detectors his anode procedure may work with. Frank just offhandedly mentions neutron tubes. Again, Frank does not really apply. The addition of air, moisture, oxygen into these neutron detectors would be disastrous to the detector. You can have no electronegative gases. G.F. Knoll “Radiation Detection and Measurement”. 1979.

Further, Frank teaches away from neutron detectors of this type by virtue of adding electronegative gas components to these devices. Adding water, moisture, Oxygen or Air to these neutron detectors would instantly render them useless. By support, an extract from Glenn F. Knoll’s text “Radiation Detection and Measurement” at Page 534:

with a diameter as large as possible so that most neutron interactions occur far away from the wall. Another is to increase the pressure of the ^3He gas to reduce the range of the charged particle reaction products. Because of the low atomic mass of ^3He , the ranges of the reaction products are unusually long and the wall effect is considerably more significant than for a BF_3 tube of the same size and fill gas pressure. One method of reducing the charged particle ranges is to add a small amount of a heavier gas to the ^3He to provide an enhanced stopping power. A detailed analysis of the wall effect in ^3He counters can be found in Ref. 22.

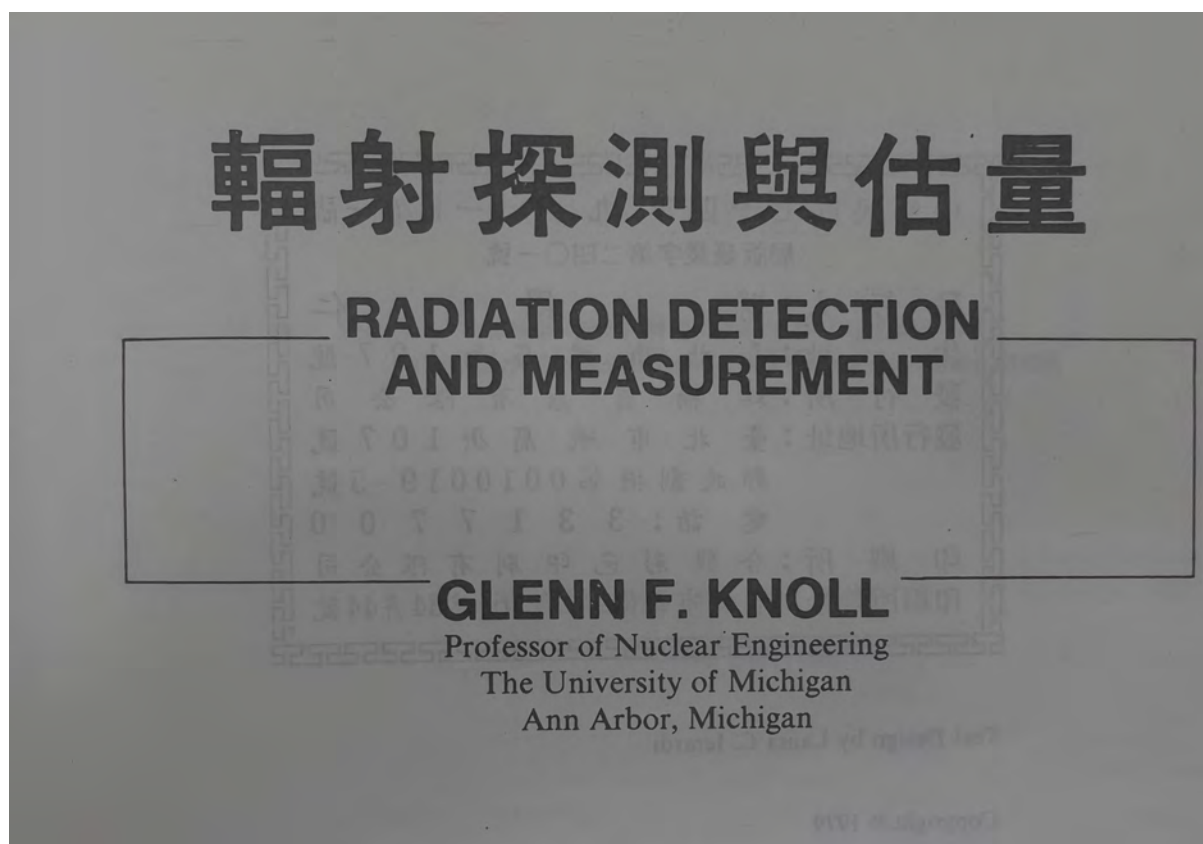
Compared with BF_3 tubes, ^3He counters can be operated at much higher pressures with acceptable gas multiplication behavior, and are therefore preferred for those applications in which maximum detection efficiency is important. The lower Q -value of the ^3He reaction, however, makes gamma ray discrimination more difficult than for an equivalent BF_3 tube.

As with all proportional counters, purity of the gas is critical, and the most typical cause of failure is leakage of air into the tube over long periods of time. Another factor is the buildup of electronegative poisons in the gas with use. As in BF_3 tubes, a layer of activated charcoal within the tube has been shown to be effective in removing these poisons and can extend the useful lifetime of a ^3He detector²³.

C. Fission Counters

The fission reaction can also serve as a means of converting a slow neutron into ionizing reaction products which can then be detected by conventional means. One outstanding characteristic of the fission reaction is the large amount of energy (200 MeV) liberated in the reaction, about 160 MeV of which appears as

This section indicates no oxygen, air, moisture in a BF_3 or ^3He tube. The title page follows:



Zhou does not overcome the shortcomings of Frank. Regarding Zhou, this reference is generally about increasing the ionization efficiency with an anode structure design with ionization after the neutron absorbing event. However, Zhou does not teach or suggest the amended Claim 1 feature of the interior of the at least one gas tube is coated with a predetermined mass thickness of 0.75-1.25 mg/cm² in Boron 10; wherein the detector gas contains a predetermined mixture of Neon, Argon, a reduced amount of a neutron absorbing gas component including partial pressures of at least one of ³He and BF₃ and a quench component; and whereas the argon, neon and the quench component increase the total pressure and adjust the waveform of the detector output to form for measurement.

In Paragraph [0057] of Zhou, "... the detector includes fill gas that produces electrons and ions in response to ionizing charged particles that travel through the gas chamber" generated from the conversion layer. Zhou goes on to discuss his *serpentine conductive pathway* which induces production of further electrons (avalanche) inducing the electrical pulse; through a long meandering anode.

In Paragraph [0058] of Zhou, He3 and BF3 are described in the context of regular detectors: “*made of metals with low Z*” and *a fill gas with high Z meaning He3 or BF3* based detectors. Again, Zhou is describing previous detectors with standard materials.

Paragraph [0059] still talking about the physics and mechanical design of a general detector. Selection of the cathode material for mechanical rigidity, and a low neutron capture cross section of the cathode material. Zhou is setting up later in [59] that if you add a conversion layer, if it's too thick it will reduce the efficiency. The charged particles cannot escape from the coating.

Paragraph [0060] Zhou is talking about conversion layers, and states as he shows He3 in table 1 only as a comparison to the coating parameters. It shows He3 gas tube has better stats.

Paragraph [0062] of Zhou leads to the discussion of optimum thicknesses of the conversion coating. But nowhere does Zhou suggest that He3 and or BF3 could be added to boost efficiency in excess of a coating. the table clearly shows that He3 is a great medium.... Large cross section, its just not as available these days and is very expensive.

Paragraphs [0065] - [0070] this is getting into the point of Zhou, the anode. These paragraphs discuss the ionization physics and impact ionization, the impact caused by the acceleration of charged particles after the neutron capture in the optimum thickness. There is not discussion of increasing the neutron capture efficiency by the addition of neutron absorbing gasses in conjunction with the conversion coating. Actually there is really nothing in the Zhou patent to lead one to adding a neutron absorbing gas to make both an active surface and an active volume. It gets to that “long felt need and failure to resolve this need” that was never explored, maybe then its not as it may first appear.

In Paragraph [0074] of Zhou, there is the discussion of the physics of the quench gas and pulse dynamics. How the pulse amplitude and duration depend on the fill gas and the magnitude of the E field produced between the patented anode and cathode. Dead time is discussed, which has to do with the recovery and resolution of the pulse, again about the fill gasses type and pressures. Quench prevents avalanche process from long durations of time recovery.

In Paragraph [75] of Zhou, there is more discussion about the fill gas, Argon is again the main medium and its discussion with the anode.

Hence, no stated discussion of further using neutron reacting volumetric gases with the surface coatings to improve efficiency. The fill gases in Zhou are really defined with respect to the anode design and the secondary electrons and ions produced in the ArNe, quench for the ionization process. This is getting a better pulse from each single neutron event.

Zhou remains silent throughout in any discussion of adding neutron absorbing gas to increase the efficiency of the detector. Basically, Zhou teaches away from this by never discussing those attributes.

Lacy does not overcome the shortcomings of Frank and Zhou. Lacy provide a standard discussion about straw tubes. These detectors add surface area by adding more surface in the form of “straw” like shapes interior coated in something like B10, then add a fill gas such as the Nobel gasses Ar, and Ne, *etc.* and a quench. At Paragraph [0011] of Lacy is specifically a discussion about **high flux** detectors. Efficiency of the detector is not so important if you have a big source of radiation like neutrons. Here Lacy fill gas is specifically Ar, Ne, ethane and therefore teaches away from adding neutron absorbing gases by never mentioning it. *High efficiency is not needed in an area with high neutron emission.*

At Paragraph [0016], Lacy teaches away from adding volumetric detection via neutron absorbing gas fill by stating that He3 cannot achieve the high-rate operation. Not without extreme pressures and expense. When you have low neutron emissions, it is very important to increase the counting statistics and measured spectrum. Hence, you either measure for a much longer time or build a much more efficient detector. I have to choose the more effective detector. This is the **hybrid surface and volumetric absorbing** media detector in my specification.

Paragraph [0014] of Lacy does also discuss optimum coating thicknesses due to limited charged particle escape energy. Further, Paragraphs [0015], [0055] describes flow of counter gas ... Ar and ethane... non-neutron absorbing of course. A flowing gas detector is not a single tube or single static detector.

The other cited references do not overcome the shortcomings of Frank, Zhou, and Lacy. Particularly, these references, either alone or in combination, do not teach or fairly suggest the

amended Claim 1 features of the interior of the at least one gas tube is coated with a predetermined mass thickness of 0.75-1.25 mg/cm² in Boron 10; wherein the detector gas contains a predetermined mixture of Neon, Argon, a reduced amount of a neutron absorbing gas component including partial pressures of at least one of ³He and BF₃ and a quench component; and whereas the argon, neon and the quench component increase the total pressure and adjust the waveform of the detector output to form for measurement.

In view of the foregoing, the cited references, either alone or in combination, do not teach or suggest the Claim 1 features of the interior of the at least one gas tube is coated with a predetermined mass thickness of 0.75-1.25 mg/cm² in Boron 10; wherein the detector gas contains a predetermined mixture of Neon, Argon, a reduced amount of a neutron absorbing gas component including partial pressures of at least one of ³He and BF₃ and a quench component; and whereas the argon, neon and the quench component increase the total pressure and adjust the waveform of the detector output to form for measurement. The other independent claims have the same or similar features. Accordingly, it is respectfully submitted that the rejections of the claims under 35 U.S.C. § 103 should be withdrawn and the claims allowed at this time.

V. Conclusion

In view of the foregoing, it is respectfully submitted that this application be placed into condition for allowance. Favorable consideration and prompt allowance of the claims are earnestly solicited.

Should the Examiner believe that anything further would be desirable in order to place this application in even better condition for allowance, the Examiner is invited to contact the undersigned at the telephone number set forth below.

Deposit Account

The Commissioner is hereby authorized to charge any fees associated with the filing of this correspondence to Deposit Account No. 50-5046.

ATTORNEY REFERENCE NO.: 202-11-REI
Application No.: 17/167,404

PATENT

Date: November 10, 2025

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